NOAA_data

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyr)
library(VIM)
```

```
## Loading required package: colorspace

## Loading required package: grid

## VIM is ready to use.

## Suggestions and bug-reports can be submitted at: https://github.com/statistikat/VIM/issues

##
## Attaching package: 'VIM'

## The following object is masked from 'package:datasets':
##
##     sleep
```

```r
library(mapview)
library(sf)
```

```
## Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
```

```r
library(mice)
```

```
##
## Attaching package: 'mice'
```

```
## The following object is masked from 'package:stats':
##
##     filter

## The following objects are masked from 'package:base':
##
##     cbind, rbind
```

```
library("geosphere")
library("fields")
```

```
## Loading required package: spam

## Spam version 2.11-1 (2025-01-20) is loaded.
## Type 'help( Spam)' or 'demo( spam)' for a short introduction
## and overview of this package.
## Help for individual functions is also obtained by adding the
## suffix '.spam' to the function name, e.g. 'help( chol.spam)'.

##
## Attaching package: 'spam'

## The following objects are masked from 'package:base':
##
##     backsolve, forwardsolve

## Loading required package: viridisLite

##
## Try help(fields) to get started.
```

##Import (However)

```
X <- read.csv("/Users/chrisduran/Downloads/4066002.csv")
Weather <- X
```

```
length(unique(Weather$STATION))
```

```
## [1] 507
```

```
weather_subset <- Weather %>%
  select(TMAX, TMIN, ELEVATION)
```

```
NOAA_computed <- mice(weather_subset, m =5 , method ='pmm', seed =500)
```

```
##
##  iter imp variable
##   1   1  TMAX  TMIN  ELEVATION
##   1   2  TMAX  TMIN  ELEVATION
```

```
##    1   3  TMAX  TMIN  ELEVATION
##    1   4  TMAX  TMIN  ELEVATION
##    1   5  TMAX  TMIN  ELEVATION
##    2   1  TMAX  TMIN  ELEVATION
##    2   2  TMAX  TMIN  ELEVATION
##    2   3  TMAX  TMIN  ELEVATION
##    2   4  TMAX  TMIN  ELEVATION
##    2   5  TMAX  TMIN  ELEVATION
##    3   1  TMAX  TMIN  ELEVATION
##    3   2  TMAX  TMIN  ELEVATION
##    3   3  TMAX  TMIN  ELEVATION
##    3   4  TMAX  TMIN  ELEVATION
##    3   5  TMAX  TMIN  ELEVATION
##    4   1  TMAX  TMIN  ELEVATION
##    4   2  TMAX  TMIN  ELEVATION
##    4   3  TMAX  TMIN  ELEVATION
##    4   4  TMAX  TMIN  ELEVATION
##    4   5  TMAX  TMIN  ELEVATION
##    5   1  TMAX  TMIN  ELEVATION
##    5   2  TMAX  TMIN  ELEVATION
##    5   3  TMAX  TMIN  ELEVATION
##    5   4  TMAX  TMIN  ELEVATION
##    5   5  TMAX  TMIN  ELEVATION
```

```r
complete_NOAA_Comp <- complete(NOAA_computed, 1)
```

```r
w_subset <- Weather %>%
  select(STATION, NAME, DATE, LONGITUDE, LATITUDE)

merged_data <- bind_cols(w_subset, complete_NOAA_Comp)

merged_data <- merged_data %>%
  filter(STATION != "US1CAHM0173")
```

```r
# Step 1: Get top 5 stations with most complete data
top_stations <- merged_data %>%
  group_by(STATION) %>%
  summarize(n_dates = n_distinct(DATE)) %>%
  arrange(desc(n_dates)) %>%
  slice_head(n = 20) %>%
  pull(STATION)

# Step 2: Filter to just those stations
top_data <- merged_data %>%
  filter(STATION %in% top_stations)

# Step 3: Get dates shared by all 5
shared_dates_top <- top_data %>%
  group_by(DATE) %>%
  summarize(n_stations = n_distinct(STATION)) %>%
  filter(n_stations == 20) %>%
  pull(DATE)
```

```r
# Step 4: Final subset
df_shared_top <- top_data %>%
  filter(DATE %in% shared_dates_top)
```

```r
est_mergeW <- df_shared_top %>%
  arrange(STATION, NAME, DATE, ELEVATION) %>%
  group_by(STATION, NAME, ELEVATION) %>%
  pivot_wider(
    names_from = DATE,
    values_from = c(TMAX, TMIN),
    names_sep = "_"
  ) %>%
  ungroup()
```

```r
new_estW <- est_mergeW %>%
  filter(STATION != "US1CAHM0173")
```

```r
noaa_new <- new_estW %>%
  select(-NAME, -STATION)

write.csv(noaa_new, file= "NOAA_increased.csv")
```